

HSINCHU CITY, Taiwan

WMO: 467570

Lat: 24.8278N

Lon: 121.0142E

Elev: 88

StdP: 14.65

Time Zone: 8.00 (E08)

Period: 90-98

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	48.4	50.6	38.7	34.7	51.6	41.5	38.6	54.2	24.2	56.8	22.2	56.5	10.1	50	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	91.7	80.0	90.6	79.8	89.4	79.5	81.7	89.0	80.9	88.3	80.2	87.6	8.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
79.7	154.6	86.3	78.8	150.2	85.7	78.1	146.6	85.2	45.6	88.9	44.7	88.9	44.0	87.8	85.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	20.7	18.3	41.2	96.8	5.8	4.0	37.1	99.7	33.7	102.1	30.5	104.4	26.3	107.3
			39.9	83.2	4.8	1.3	36.5	84.1	33.6	84.9	30.9	85.6	27.4	86.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	72.6	60.0	60.3	64.3	70.9	76.6	81.1	83.8	83.4	80.1	74.8	70.6	64.8	(6)
(7)		DBStd	9.40	4.83	5.00	5.63	6.01	4.61	3.77	2.04	2.04	3.32	3.95	4.35	4.57	(7)
(8)		HDD50	4	3	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD65	480	164	149	81	20	1	0	0	0	0	0	8	57	(9)
(10)		CDD50	8262	312	287	445	628	823	932	1047	1036	904	769	619	460	(10)
(11)		CDD65	3262	8	16	61	198	359	482	582	571	454	304	177	50	(11)
(12)		CDH74	29884	7	25	170	1028	2813	5286	7226	6731	4312	1620	596	70	(12)
(13)		CDH80	10204	0	0	16	151	725	1950	3086	2616	1307	276	74	3	(13)
(14)	Wind	WSAvg	7.1	8.1	7.5	6.4	5.6	4.7	6.1	5.6	5.4	6.8	9.9	9.1	9.5	(14)
(15)	Precipitation	PrecAvg	94.6	5.0	5.9	5.5	5.9	9.1	9.5	7.5	9.8	13.0	11.1	8.1	5.4	(15)
(16)		PrecMax	154.8	12.2	14.3	9.1	18.1	20.5	18.7	16.2	23.0	43.8	46.4	20.5	13.1	(16)
(17)		PrecMin	50.9	1.8	1.1	1.9	1.1	2.2	3.0	1.3	2.4	2.4	1.6	1.3	0.6	(17)
(18)		PrecStd	23.0	2.8	3.5	1.6	3.7	4.8	3.9	4.4	6.0	9.0	9.1	5.3	3.4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	74.8	77.3	82.3	86.1	90.9	91.4	93.6	93.6	92.4	87.6	85.1	79.2	(19)
(20)			MCWB	64.4	68.1	72.9	75.4	79.5	79.5	79.9	80.0	79.1	76.7	74.5	69.9	(20)
(21)		2%	DB	71.5	73.7	78.3	83.7	88.2	89.8	91.7	91.5	89.9	85.0	82.2	75.8	(21)
(22)			MCWB	63.2	66.2	70.8	74.8	78.4	79.9	80.2	80.3	78.6	75.5	72.0	68.2	(22)
(23)		5%	DB	69.0	70.9	75.4	81.8	85.7	88.7	90.5	90.2	88.2	83.1	80.0	73.2	(23)
(24)			MCWB	62.2	64.3	68.9	74.1	77.3	79.4	79.9	79.9	78.1	74.2	71.1	66.6	(24)
(25)		10%	DB	66.5	67.7	72.8	79.7	83.9	87.6	89.3	88.9	86.4	81.1	77.6	71.4	(25)
(26)			MCWB	61.1	62.3	67.6	73.3	76.2	78.7	79.5	79.4	77.6	73.1	69.9	65.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.4	70.3	73.8	77.2	81.0	82.9	82.7	82.7	81.0	79.0	75.3	72.1	(27)
(28)			MCDWB	71.4	75.2	79.0	83.8	88.8	87.9	90.2	90.0	89.2	85.5	81.9	77.2	(28)
(29)		2%	WB	64.8	67.1	71.7	75.9	79.1	81.0	81.6	81.5	80.0	77.3	73.9	70.1	(29)
(30)			MCDWB	69.4	72.7	77.2	81.9	86.2	87.5	89.4	88.9	87.7	82.6	79.9	73.9	(30)
(31)		5%	WB	63.2	64.9	70.0	74.9	77.9	80.1	80.8	80.7	79.2	76.0	72.5	68.1	(31)
(32)			MCDWB	67.5	69.6	74.9	80.6	84.4	86.9	88.5	88.2	86.1	81.1	78.2	72.1	(32)
(33)		10%	WB	61.9	62.9	68.2	73.7	77.0	79.2	80.1	80.0	78.3	74.7	71.1	66.4	(33)
(34)			MCDWB	65.7	67.0	71.8	79.1	83.1	85.9	87.6	87.1	84.6	79.6	76.5	70.5	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.5	9.0	9.5	10.1	10.6	10.1	11.3	10.8	10.9	9.8	10.8	9.7	(35)
(36)			MCDBR	14.3	14.6	14.4	13.0	12.9	11.6	12.4	12.8	13.6	13.0	14.7	13.5	(36)
(37)			MCWBR	8.6	8.7	7.9	6.8	6.8	6.2	6.1	5.8	6.6	7.1	7.6	8.1	(37)
(38)		5% WB	MCDBR	12.1	13.0	13.3	11.7	12.1	11.0	11.9	11.7	12.1	10.7	13.2	11.2	(38)
(39)			MCWBR	8.0	8.1	7.8	6.8	7.0	6.3	6.3	6.0	6.2	6.5	7.7	7.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.542	0.588	0.701	0.706	0.603	0.559	0.552	0.574	0.554	0.541	0.532	0.538	(40)	
(41)		taud	1.880	1.802	1.598	1.638	1.900	2.032	2.046	1.979	2.012	1.991	1.988	1.906	(41)	
(42)		Ebn at Noon	222	222	205	209	232	240	241	236	238	233	224	216	(42)	
(43)		Edh at Noon	57	65	84	82	63	55	54	58	55	53	50	53	(43)	
(44)	All-Sky Solar Radiation	RadAvg	770	863	1050	1176	1366	1564	1892	1715	1494	1229	885	763	(44)	
(45)		RadStd	140	154	135	171	234	184	112	158	157	137	103	119	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (20 neighbors)	+0.45	N/A	N/A	+0.59	N/A	N/A	N/A	+221	+175	

Nomenclature: See separate page